Evaluate yourself - 5

At a particular temperature, the K_w of a neutral solution was equal to 4×10^{-14}. Calculate the concentration of $[H_3O^+]$ and $[OH^-]$.

8.4 The pH scale

We usually deal with acid / base solution in the concentration range 10^{-1} to 10^{-7}M. To express the strength of such low concentrations, Sorensen introduced a logarithmic scale known as the pH scale. The term pH is derived from the French word '*Purissance de hydrogene*' meaning, the power of hydrogen. pH of a solution is defined as the negative logarithm of base 10 of the molar concentration of the hydronium ions present in the solution.

$$pH = - \log_{10}[H_3O^+] \qquad .....(8.5)$$

The concentration of H_3O^+ in a solution of known pH can be calculated using the following expression.

$$[H_3O^+]=10^{-pH} \text{ (or) } [H_3O^+]= \text{antilog of (-pH)} \qquad .....(8.6)$$

Similarly, pOH can also be defined as follows

$$pOH = - \log_{10}[OH^-] \qquad .....(8.7)$$

As discussed earlier, in neutral solutions, the concentration of $[H_3O^+]$ as well as $[OH^+]$ is equal to 1×10^{-7}M at 25°C. The pH of a neutral solution can be calculated by substituting this H_3O^+concentration in the expression (8.5)

$$pH = - \log_{10}[H_3O^+]$$

$$= - \log_{10} 10^{-7}$$

$$=(-7)(-1)\log_{10}10= +7\ (1)= 7 \qquad [\because \log_{10}10=1]$$

Similary, we can calculate the pOH of a neutral solution using the expression (8.7), it is also equal to 7.

The negative sign in the expression (8.5) indicates that when the concentration of $[H_3O^+]$ increases the pH value decreases. For example, if the $[H_3O^+]$ increases from to 10^{-7}to10^{-5}M, the pH value of the solution decreases from 7 to 5. We know that in acidic solution, $[H_3O^+] > [OH^-]$, i.e., $[H_3O^+] > 10^{-7}$. Similarly in basic solution $[H_3O^+] < 10^{-7}$. So, we can conclude that acidic solution should have pH value less than 7 and basic solution should have pH value greater than 7.

8.4.1 Relation between pH and pOH

A relation between pH and pOH can be established using their following definitions

$$pH=-\log_{10}[H_3O^+] \qquad .....(8.5)$$

$$pOH = - \log_{10}[OH^-] \qquad .....(8.7)$$

Adding equation (8.5) and (8.7)

XII U8-Ionic equilibrium.indd 9 16-12-2022 19:33:42